

Tactile and Directional Intent

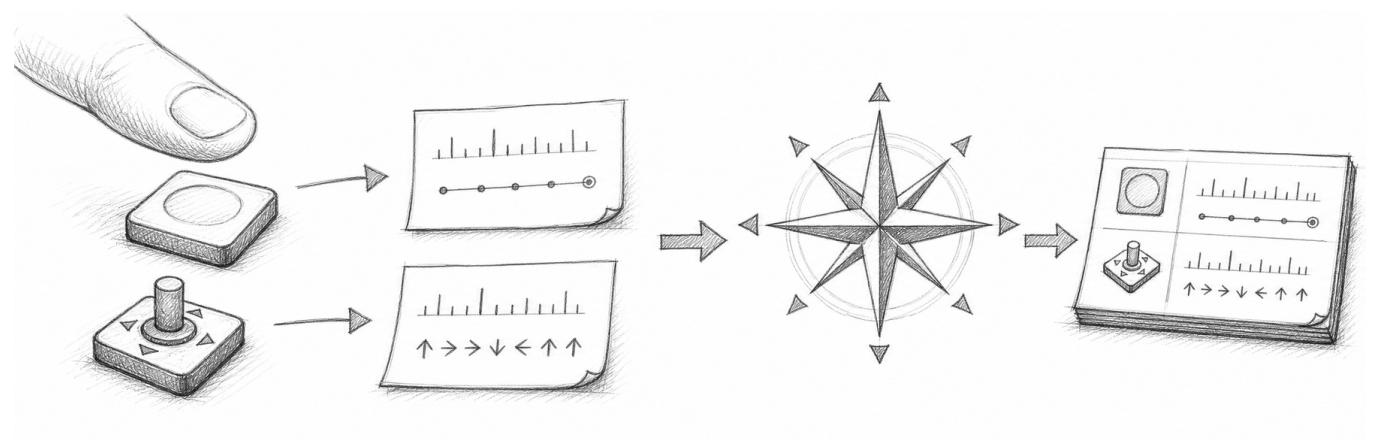
Turn two copied observations into one coherent decision

E0 COPIED-EVIDENCE REPLAY — NO POWERED INPUT CLAIM

Lesson	046 — component	Time	90–120 minutes
Prerequisites	Lessons 031 and 037	Board	compile-only Mega replay
Evidence	named memory result cells	Boundary	no GPIO, ADC, bus, or live clock

Responsibility. Qualify one copied contact action and one copied directional observation, then publish one age-qualified interaction snapshot. Preserve both source identities, sequences, statuses, and ages. Do not infer capacitive sensing, gesture, proximity, pulse, or identity.

1 Copy evidence before deriving intent

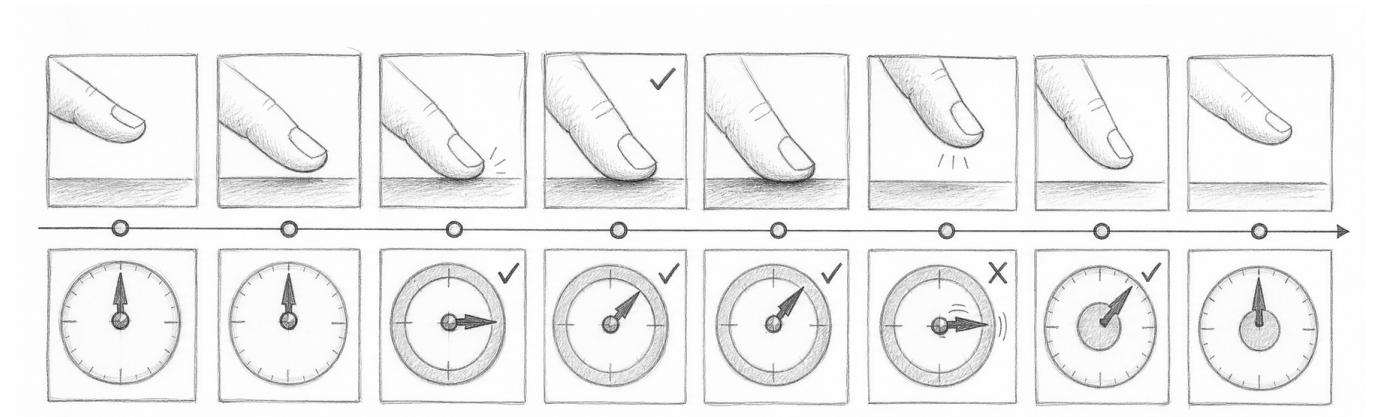


Alternative text: a graphite pencil drawing follows an inert touch token and directional token into separate copied evidence cards, an eight-way compass, and one bundled interaction snapshot; no wires or powered hardware appear.

`InteractionIntentPolicy` is a narrow facade, not a universal sensor interface. It privately reuses `Contact Dynamics` and derives direction from already-calibrated axes in permille. The source domain is kind, source ID, and configuration revision.

Copied fact	Why it remains visible
two source domains	prevents evidence from different producers being confused
two timestamps and ages	prevents a fresh joystick from hiding stale contact
two sequences	exposes repeats, gaps, regressions, and source changes
two producer statuses	prevents either fault from being hidden by the other
saturation marker	preserves valid extreme directional evidence

2 Qualify contact; apply directional hysteresis



Alternative text: a pencil timeline shows brief contact noise, qualified contact and release above; below, copied direction moves from neutral beyond an engage ring, through octants, then inside a smaller release ring. Checks mark admitted transitions and a cross marks rejected noise.

Use the canonical configuration: active-high contact; qualify and release 8 ms; refractory 80 ms; stuck-active 2000 ms; maximum contact and directional ages 100 ms; directional engage 300 permille and release 200 permille.

Rule	Predicted result
brief active evidence	no qualified touch event
qualified active evidence	one <code>touchEvent</code>
engaged direction changes octant	one <code>directionEvent</code>
magnitude stays between 200 and 300	retain the previously engaged sector
magnitude falls below 200	publish <code>Neutral</code> and one event
exact copied replay	recompute age; clear one-update events

Magnitude is $\max(|x|, |y|)$ using widened arithmetic. For non-neutral axes, let *major* be the larger absolute axis and *minor* the smaller. When $\text{minor} \times 1000 \leq \text{major} \times 414$, choose the principal axis; exact equality belongs to that axis. Otherwise choose the diagonal. Equal absolute axes are diagonal.

3 Octant worksheet

<i>x</i>	<i>y</i>	magnitude	predicted direction / event
600	0	_____	_____
600	248	_____	_____
600	249	_____	_____
-600	-600	_____	_____
100	100	_____	_____

4 Replay the eight named frames

Use the copied fixture table in the Lesson 046 compile-only sketch. Predict before inspecting each volatile memory cell. The sketch follows acquire, configure, start; then observe, decide, actuate. “Actuate” writes only an in-memory mirror.

Frame	Copied evidence	Prediction	Observed result cell / interpretation
1	low; neutral	current baseline	_____
2	high at 1 ms	not yet qualified	_____
3	high at 9 ms; 600, 0	touch event; East	_____
4	high; 600, 600	NorthEast event	_____
5	1000, 1000, saturated	NorthEast; saturation retained	_____
6	low; 100, 100	release pending	_____
7	low at 20 ms; neutral	touch release	_____
8	copied directional HardwareFailure status	source fault; no usable event	_____

5 Preview, validate, commit

1. `preview(...)` validates chronology, source domains, statuses, ranges, and semantic repeats without changing committed state.
2. `canCommit(candidate)` confirms owner and generation binding. A foreign, stale, reset-era, or already-used preview cannot commit.
3. `commit(candidate)` advances the private contact policy once, then publishes one complete interaction snapshot.

Atomicity prediction. Change one axis while keeping the same source-domain sequence. Preview must reject it. Every field in the last committed snapshot must remain unchanged. Repeat with a future timestamp, regression, and ambiguous half-range sequence delta.

6 Status and quality precedence

For admitted evidence, source fault outranks stuck-active, which outranks stale, which outranks current. Structural chronology faults are not published as **TimingFault**; they reject before commit. The enum value is reserved for ABI stability.

7 Diagnose without inventing hardware

Case	Prediction	Observation / interpretation
contact older than 100 ms	stale	_____
direction older than 100 ms	stale	_____
axis outside $[-1000, 1000]$	reject atomically	_____
valid saturated extreme	current, flag retained	_____
same sequence, changed payload	reject atomically	_____
new source domain	fresh direction baseline	_____
stuck active contact	stuck-active dominates	_____

8 Separate acquisition from safe state

Stage	Predict	Observe and interpret
Acquire	validate configuration; initialize pure policy	status cell: _____
Observe	admit one complete copied frame	two sequences and quality: _____
Reset	clear history and one-update events	canonical invalid snapshot: _____

This E0 component owns no electrical resource, so initialization proves only software acquisition and reset proves only software safe state. Neither establishes an input voltage, inactive pin level, debounce circuit, test point, or power-removal behavior.

9 Resource record

Gate	Record
<code>sizeof(InteractionIntentPolicy)</code>	measured: _____ bytes; target / hard ceiling: 192 / 256 bytes
complete Mega sketch	flash: 6956 bytes (limit 16384); static SRAM: 733 bytes (limit 2048)
owned resources	no heap, pins, timers, interrupts, buses, ADC, or live clock
host evidence	deterministic, exception-unwind, sanitizer, and size gates: _____

Do not fill measured cells from estimates. Copy values from the promoted size gate for the tested commit.

10 Blank E1 input acceptance record

Leave every field blank until separately qualified powered work is performed and reviewed. E1 still authorizes no stepper coil or moving load.

Required field	Bench record
board revision / tested commit	_____
exact touch/contact specimen and markings	_____
exact directional specimen and markings	_____
primary-source electrical references	_____
supply / output levels / polarity	_____
instrument / tool version	_____
resource-acquisition prediction	_____
resource-acquisition observation	_____
non-Serial test point prediction / observation	_____
safe-state prediction / observation	_____
power-removal result	_____
interpretation / reviewer / date	_____

11 Claim boundary

A “Metal Touch” retail name does not establish capacitive behavior, active polarity, supply, output level, debounce, or electrical safety. Copied joystick axes are not raw ADC readings and do not make this policy an **AnalogJoystick** owner. Nothing here detects a heartbeat, pulse, gesture, proximity, or person. E2 qualification of the exact stepper, driver, supply, restraint, stop, and power removal belongs to Lessons 047–048 and remains open.

12 Evidence links

- Public interface
- Deterministic host tests
- Compile-only copied-evidence replay
- Reviewed design and open E1/E2 gates

The searchable HTML lesson is the hardware-independent accessible route. This untagged print edition makes no PDF/UA, powered-input, motion, or physical acceptance claim.